

AMAR MAHATO

AI Engineer | LLM & RAG Systems

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PROFILE

I build LLM and RAG features and the backends that serve them, and I focus on getting the retrieval and grounding right so the answers can be trusted.

EXPERIENCE

IntelGrader

Remote

AI Engineer Intern

May 2026 – Present

- Trained RF-DETR object detectors for the grading pipeline: one to flag scratches and struck-out (cancelled) handwriting so it is masked before OCR, another to detect printed diagrams and tables. Set per-label masking thresholds, tuned through repeated experiments so the model almost never hides valid text (it would rather miss a scribble than mask a real answer), plus a promotion gate that automatically rejected a model version that failed the bar.
- Built an adversarial verification tool for the OCR stage that caught a systematic normalization bug and blocked a flawed 6,000-page run before it shipped, and ran comparisons across candidate OCR models to see which performed best.
- Built the team's internal tooling end to end: a real-time annotation and review platform used to verify 8,000+ images (three build generations, live multi-user sync with edit locks), a GPU-sharing coordinator with live telemetry, and an on-device Android auto-scanner that detects and captures pages (Kotlin, CameraX, ONNX).

PROJECTS

Concize | Node.js, WebSockets, FunASR/CAM++, pyannote, Qdrant

[GitHub](#)

- A real-time assistant for online meetings that transcribes the call live, labels who is speaking, and answers questions using only what was actually said, so it never makes things up.
- Built the backend and live audio pipeline on Node.js and WebSockets: audio streams in and is transcribed and split into per-speaker turns in real time, each stored so it can be searched and cited later.
- Ran speaker identification with a self-hosted model (FunASR + CAM++) instead of a managed API, caught it silently merging speakers, and lifted accuracy from 85% to 94% on a 40-speaker benchmark by tuning the threshold; overlaps are detected with pyannote and left unattributed.
- Built the retrieval and grounding: hybrid search (dense + keyword) pulls the right slice of a long meeting, and every answer cites the exact transcript turn or admits it doesn't know.

Trecl | Python, LangGraph, multi-agent, Qdrant

[GitHub](#)

- A research tool for job-seekers that studies a company across its website, GitHub, and open roles, then drafts a personalized cold email and project pitch for each opportunity worth chasing.
- Built as six specialized agents in a LangGraph pipeline: one gathers and indexes company info, two scan job boards and the company's GitHub in parallel for real openings and fixable issues, and the rest rank everything into tiers and write the outreach, pausing for the user to choose first.
- Built the GitHub agent so it can't invent inputs (it verifies a repo or label exists before using it, and strips injected search operators), and kept it dependable under API limits with round-robin key rotation across four services.

IcBerg | Python, FastAPI, sandboxed execution, Qdrant

[GitHub](#)

- A chat assistant for data analysis: you ask questions in plain English and it computes the answers and charts for you, no coding required.
- The AI writes small analysis snippets instead of touching the data directly; each is statically scanned to block dangerous operations, then run in an isolated process with hard memory and time limits, so generated code can never hang or crash the server.
- Built on FastAPI with a reasoning agent loop, automatic failover to a second LLM provider, and a semantic cache that reuses past answers when a new question means the same thing, all backed by 180 tests.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, SQL • **Applied AI:** RAG (hybrid retrieval, reranking, contextual chunking), LLM agents, LLM/model evaluation, prompt-injection defense, real-time speech-to-text & diarization

Backend: FastAPI, Node.js, Express.js, WebSockets, SSE, REST APIs, JWT/OAuth2 • **Data:** PostgreSQL, Supabase, Qdrant, Redis, vector search

ML: object detection & fine-tuning (RF-DETR), model evaluation, calibration & risk control • **Infra & Tools:** AWS (EC2, S3), Modal, Docker, GitHub Actions, RunPod, Cloudflare (Workers, R2), Linux, Git, pytest

EDUCATION

Dr. B. C. Roy Engineering College

Durgapur, West Bengal, India

Bachelor of Technology, Computer Science and Engineering

2023 – 2027